A High-Performance Rule-Based Multi-Agent Framework for Spatio-Temporal Vector Dynamics: Application to *Anopheles stephensi* in Ethiopia

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Abstract

Background: The invasive malaria vector *Anopheles stephensi* threatens urban Africa, but data scarcity limits conventional modelling. **Methods:** We developed a modular agent–based framework coupling a temperature–dependent Metzler matrix life–cycle model with spatial environmental layers (population, buildings, climate), behavioural rules, and a lazy spatial grid. The model was parameterised for *An. stephensi* and run for six months over two Ethiopian regions (Somali and Oromia). **Results:** Simulated populations reached near–equilibrium (Oromia $\lambda_{\max} = 1.077$; Somali $\lambda_{\max} = 1.135$) with stage durations (Oromia: larvae 15.7 d, pupae 4.8 d; Somali: larvae 11.4 d, pupae 4.3 d) matching laboratory data. Urban cores showed highest suitability (Continuous Boyce Index: Oromia 0.504; Somali 0.782). In Oromia, adult abundance correlated weakly positively with temperature ($r = 0.12, p = 0.12$) and moderately positively with precipitation ($r = 0.62, p < 0.001$). In Somali, where temperature was nearly constant, a moderate negative correlation with precipitation was observed ($r = -0.48, p < 0.001$). **Conclusion:** The framework reproduces known *An. stephensi* ecology and identifies urban hotspots. Its YAML–driven parameterisation makes it adaptable to other species and regions, supporting evidence–based vector control in data–limited settings.

Keywords: multi-agent systems, vector-borne disease, *Anopheles stephensi*, Metzler matrix, high-performance computing, spatial ecology.

1 Introduction

Vector-borne diseases (VBDs) represent one of the most profound challenges to global health security, accounting for approximately 700,000 deaths annually and imposing significant economic burdens on developing nations (Lowe and Codeço, 2025). Malaria alone threatens nearly half the world's population, with Sub-Saharan Africa bearing 94% of the global case burden as (Organization, 2024). While the past two decades witnessed substantial reductions in malaria mortality through coordinated interventions, this hard-won progress faces unprecedented threats from ecological and demographic shifts. The rapid urbanization of Sub-Saharan Africa, currently the world's fastest urbanizing region (Sulemana et al., 2019), has created novel transmission ecologies that challenges historical assumptions about vector ecology and control strategies (Joseph et al., 2024).

The primary driver of this epidemiological transition is the globalization of previously localized vectors, particularly the invasive malaria mosquito *Anopheles stephensi*. Unlike endemic African malaria vectors that predominantly breed in natural water bodies, *An. stephensi* exhibits remarkable ecological plasticity, thriving in artificial containers and man-made structures characteristic of urban environments (Hunter et al., 2025). This species shares critical adaptive traits with *Aedes aegypti*, the primary dengue vector, including container breeding, anthropophily, and exophilic resting behaviors (Hamlet et al., 2022). The consequences of this adaptation were first recorded in Djibouti, where *An. stephensi* establishment triggered a 30-fold increase in malaria cases within seven years, transforming a nation on the verge of malaria elimination into a high-transmission zone (Hamlet et al., 2022). This vector's capacity to exploit previously low-risk urban settings necessitates fundamentally new approaches to surveillance, modeling, and intervention design (Mwema et al., 2024).

A critical bottleneck in responding to this emerging threat is the pervasive "Data Gap" characterizing many resource-constrained settings where *An. stephensi* is establishing. Effective vector control relies on robust decision-support systems built upon high-quality, high-resolution entomological and epidemiological data, yet

national surveillance programs often operate with limited financial and technological resources (Lowe and Codeço, 2025). Current modeling approaches struggle to bridge the gap between theoretical elegance and practical utility: empirical models propagate measurement errors through unreliable data streams, while process-based models frequently lack the spatial granularity and computational efficiency needed for operational deployment (Lowe and Codeço, 2025). This disconnect is particularly acute for invasive vectors, where absence of historical baseline data undermines traditional statistical approaches (Pascoe et al., 2019).

To address these challenges, we propose a novel computational framework that integrates high-performance computing methodologies with ecological process modeling. Our system implements a layered and modular, rule-based Multi-Agent Simulation (MAS) architecture that harmonizes heterogeneous spatio-temporal data streams into a unified analytical environment. This framework is part of the family of ABS (Agent Based Simulation) which are simulation models where the researcher explicitly describe the decision processes of simulated actors at micro-level (Siebers and Aickelin, 2008). The proposed framework bridges Computer Science and Environmental Science by conceptualizing environmental data as dynamic information layers, including static rasters (for instance: elevation, building density, population density), time-series climate variables, and vector data (shapefiles), that interact with autonomous agent populations. We distinguish between two agent classes: "Inert Agents" representing static environmental features (urban water containers serving as breeding sites) and "Living Agents" modeling biological vectors (*An. stephensi* mosquitoes for example) with behavioral rules derived from empirical studies.

This study demonstrates how such synthetic modeling approaches can provide critical insights where field data are scarce or unreliable. Using Somali, and Oromia woredas from Ethiopia, which are regions experiencing *An. stephensi* invasion, (Zhou et al., 2024; Hawaria et al., 2023) as a proof-of-concept, we simulate 6 months of mosquito population dynamics across the whole regions. Our results illustrate how agent-based modeling can identify high-risk micro-habitats, simulate seasonal population trajectories, and potentially assist intervention strategies in data-limited settings. By making the simulation framework open-source and computationally efficient, we aim to empower public health agencies in resource-constrained regions with decision-support tools capable of addressing the unique challenges posed by urban-adapted invasive vectors.

2 Materials and Methods

2.1 Framework Architecture

2.1.1 Overview

The simulation framework follows a modular, agent-based design that integrates spatially explicit environmental data, temperature-dependent life-cycle models, and behavioural rules. It is implemented in Java and built around four core subsystems: (i) a time manager that advances simulation time in discrete ticks (15-minutes per tick) and interpolates climate data; (ii) a spatial registry that indexes agents on a lazy grid for efficient neighbourhood queries; (iii) an agent layer system that manages agents and evaluates rules in parallel; and (iv) a lifecycle model that computes temperature-driven development and survival rates. The overall architecture is illustrated in Figure 1.

2.2 Core Components

2.2.1 Agent layers

Two agent types are implemented: `LivingAgent` (representing a mosquito) and `InertAgent` (representing a breeding site). Living agents track life stage (egg, larva, pupa, adult), age, energy, gravid status, and resting behaviour. Their movement is modelled as a random walk with a step size derived from field observations of flight range (Sayeed Quraishi et al., 1966). Inert agents maintain water volume, egg count, larval count, and carrying capacity; they update their capacity dynamically based on water level, mimicking density-dependent mortality observed in larval habitats (Thomas et al., 2016). Agent layers are processed in parallel using a partition-based container (`AgentContainer`) and a rule engine that evaluates behavioural rules (e.g., feeding, oviposition, resting) in priority order (Ogunlade et al., 2023).

2.2.2 Raster layers and climate data

Static raster layers (elevation, building density, population density) are loaded from GeoTIFF files using GeoTools and stored either in memory or memory-mapped for large grids. Climate time series (2-m temperature and total precipitation) are ingested from NetCDF files (ERA5-Land reanalysis) into interpolated raster layers that perform linear temporal interpolation between hourly steps (Chabot-Couture et al., 2014; Merkord et al., 2017).

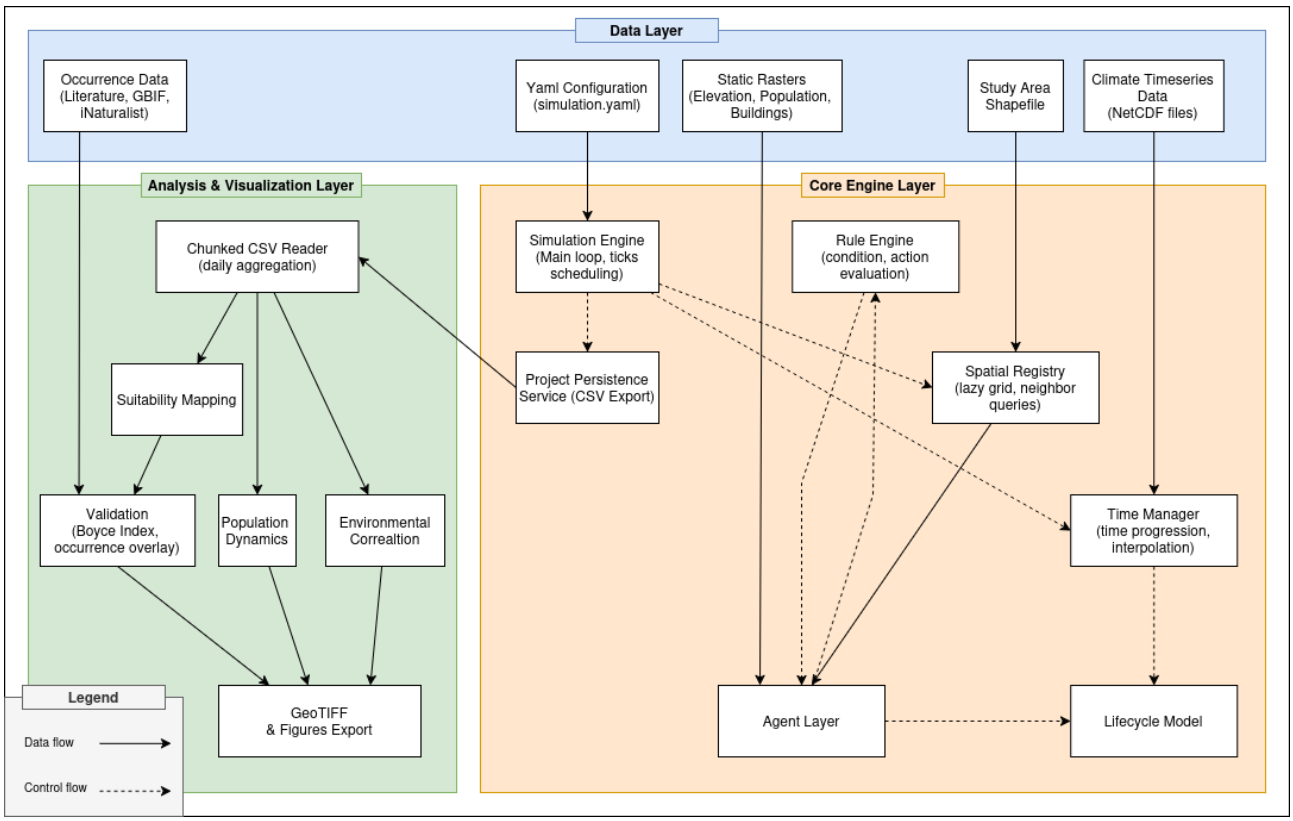


Figure 1: High-level architecture of the simulation framework. The diagram is organised into three layers: *Data Layer* (input data sources), *Core Engine Layer* (simulation components), and *Analysis & Visualization Layer* (post-processing and validation). Solid arrows denote *data flow* (e.g., configuration files, rasters, climate NetCDF, and simulation outputs). Dashed arrows denote *control flow* (e.g., tick scheduling, spatial queries, rule evaluation, and lifecycle updates). The analysis layer reads CSV snapshots generated by the persistence service and produces daily summaries, suitability maps, validation metrics, and final figures.

2.2.3 Lifecycle model

The lifecycle model follows the stage-structured discrete-time Metzler matrix framework from the work of Agboka et al. (2025). Four stages are tracked: eggs (E), larvae (L), pupae (P), and adult females (A). All rates are temperature-dependent and defined on a per-day basis. For a given daily rate r, the probability of transition or mortality over one tick of length Δt (here $\Delta t = 0.01042$ days, corresponding to 15-minute ticks) is computed via the Poisson process:

$$p = 1 - \exp(-r\Delta t).$$

The daily projection matrix is:

$$\mathbf{M}(T) = \begin{pmatrix} (1-d_E)S_E & 0 & 0 & F \cdot K \\ d_E S_E & (1-d_L)S_L & 0 & 0 \\ 0 & d_L S_L & (1-d_P)S_P & 0 \\ 0 & 0 & d_P S_P & 1-\mu_A \end{pmatrix},$$

with stage abundances evolving as $[E, L, P, A]_{t+1}^\top = \mathbf{M}(T)[E, L, P, A]_t^\top$. The temperature-dependent functions are defined as follows.

Development rates (units: day^{-1})

- **Egg and pupa development** follow the Sharpe–Schoolfield type model:

$$d_X(T) = \max\Big(0, \rho e^{\rho k} e^{-\frac{k-T}{\Delta}} + \lambda\Big),$$

  where ρ, k, Δ, and λ are species-specific parameters (see Table 1).

- **Larva development** uses the Brière model:

$$d_L(T) = \max\Big(0, aT(T - T_{\min})(T_{\max} - T)^{1/m}\Big),$$

  with a, $T_{\min}$, $T_{\max}$, and m as empirical coefficients.

Stage survival (dimensionless) Survival of eggs, larvae, and pupae is described by a Gaussian function:

$$S_X(T) = \mathrm{amp}_X \cdot \exp\left(-\frac{1}{2}\left(\frac{T-\mu_X}{\sigma_X}\right)^2\right).$$

Adult mortality (day^{-1}) Adult female mortality is temperature-dependent and modelled as an exponential quadratic:

$$\mu_A(T) = \exp(b_1 + b_2 T + b_3 T^2),$$

where b_1, b_2, b_3 are fitted coefficients.

Fecundity (eggs per female per day) Daily fecundity follows an asymmetric function:

$$F(T) = \max\left(0, ae^{bT} - \exp\left(bT_{\max} - \left[\frac{T_{\max}-T}{c}\right]^2\right)\right),$$

with parameters $a, b, T_{\max}$, and c.

Host carrying capacity Fecundity is multiplied by a dimensionless factor K that accounts for human population density (pop) and saturating effects:

$$K = 1 + \ln(1 + \mathrm{pop}).$$

Density-dependent larval mortality Each aquatic habitat (tank) has a carrying capacity cap that scales linearly with its water volume. If the number of larvae L in a tank exceeds cap, a proportion $(L - \mathrm{cap})/L$ of larvae is removed stochastically per tick, implementing a logistic-type density regulation.

All temperature-response coefficients are stored in a YAML configuration file under the `species` key. The parameter names and corresponding YAML keys are listed in Table 1.

Table 1: Generic temperature-response parameters for the mosquito lifecycle model.

Parameter	Meaning	Unit	YAML key
Egg and pupa development (Sharpe–Schoolfield)			
ρ	Development rate scale	day^{-1}	`egg_dev_rho`, `pupa_dev_rho`
k	Optimal temperature	°C	`egg_dev_k`, `pupa_dev_k`
Δ	Temperature sensitivity	°C	`egg_dev_Delta`, `pupa_dev_Delta`
λ	Baseline development offset	day^{-1}	`egg_dev_lambda`, `pupa_dev_lambda`
Larva development (Brière)			
a	Scaling factor	$\mathrm{day}^{-1}\,{}^\circ\mathrm{C}^{-(1+1/m)}$	`larva_dev_a`
$T_{\min}$	Minimum temperature	°C	`larva_dev_Tmin`
$T_{\max}$	Maximum temperature	°C	`larva_dev_Tmax`
m	Shape exponent	—	`larva_dev_m`
Stage survival (Gaussian)			
amp_X	Maximum survival	—	`<Stage>_survival_amp`
μ_X	Optimum temperature	°C	`<Stage>_survival_mean`
σ_X	Temperature tolerance width	°C	`<Stage>_survival_sigma`
Adult mortality (exponential quadratic)			
b_1	Intercept	—	`adult_mort_b1`
b_2	Linear coefficient	$^\circ\mathrm{C}^{-1}$	`adult_mort_b2`
b_3	Quadratic coefficient	$^\circ\mathrm{C}^{-2}$	`adult_mort_b3`
Fecundity			
a	Maximum fecundity scale	eggs/fem/day	`fecundity_rmax`
b	Exponential coefficient	$^\circ\mathrm{C}^{-1}$	`fecundity_c`
$T_{\max}$	Temp. of max fecundity	°C	`fecundity_Topt`
c	Quadratic decay parameter	$^\circ\mathrm{C}^{-2}$	`fecundity_c`

2.2.4 Rule engine and spatial registry

Behavioural rules (e.g., "`feed`", "`lay_eggs`", "`rest_in_building`") are defined in a YAML configuration file and evaluated using a JavaScript engine (GraalVM (Vergilio et al., 2023)). The spatial registry implements a lazy grid with a configurable cell size (default $0.001° \approx 110$ m). Neighbour queries are performed by checking cells within a Chebyshev distance $\lceil \text{radius}/\text{cellSize} \rceil$; results are cached for one second per cell to reduce computational cost.

2.2.5 Simulation Configuration

All simulation parameters are defined in a single YAML file (`simulation.yaml`). The file specifies:

- **Time settings**: start date, total number of ticks, tick duration (15-minutes).
- **File paths**: study area shapefile, elevation, building density, population rasters, and the climate NetCDF.
- **Species parameters**: coefficients for development, survival, fecundity, and adult mortality.
- **Tokens**: mapping between environment variable names (e.g., "temperature") and raster layers.
- **Agent layers and rules**: each layer contains a list of rules with a condition, an action, and a priority.
- **Seeding**: number of breeding sites and mosquitoes, habitat grid resolution, and whether to seed uniformly or using a weighted habitat suitability map.
- **Project defaults**: agent search radius, step size, maximum age.

At startup, the simulator loads the configuration, initialises the time manager, loads rasters and climate data, creates the spatial registry, builds agent layers, seeds the initial population, and starts the simulation engine.

2.2.6 Result Analysis

Simulation outputs are exported as CSV snapshots every 100 ticks. Each snapshot contains per-agent records with tick count, agent ID, layer, type, coordinates, alive flag, age, stage, energy, gravid status, and for inert agents, egg/larval counts and water volume. Environmental values (temperature, precipitation, etc) are attached to each agent's row via bilinear interpolation from the current climate frame.

The analysis, visulization and validation of the resutls are realized through a jupyter notebook (Python) which performs the following steps:

1. **Daily aggregation**: The CSV is read in chunks (100 000 rows) to compute daily adult, larva, and pupa counts, as well as breading sites statistics (mean water volume, total eggs/larvae), using the tick-to-day conversion (96 ticks per day by default).
2. **Population dynamics**: Stage counts are plotted, and stage durations are estimated from cross-correlation of smoothed population peaks. Theoretical durations are computed from species parameters at the mean observed temperature.
3. **Environmental correlation**: Pearson correlation between daily adult counts and mean temperature / precipitation is calculated.
4. **Suitability mapping**: A kernel density estimate (KDE) of adult points (bandwidth $0.002° \approx 220m$) is generated and exported as a GeoTIFF. A breeding site potential raster is created by rasterising the sum of eggs and larvae in water tanks on the last tick.
5. **Validation**: The continuous Boyce index (Hirzel et al., 2006) and the proportion of occurrence points falling into the top 20% of the suitability surface are computed using independently collected occurrence data (from GBIF, iNaturalist, literature, clipped to the study area).
6. **Export**: Daily summaries and a sample of raw data are saved for further analysis.

All figures and rasters are stored in a user-defined output directory.

2.3 Case Study Design

2.3.1 Vector and Study Site

The case study is set in Ethiopia, focusing on two contrasting regions: Somali and Oromia. Both areas have experienced recent incursions of the Asian malaria vector *Anopheles stephensi* and are considered high-risk zones for urban and peri-urban malaria transmission (Balkew et al., 2020, 2021; Hawaria et al., 2023).

Somali (6.6612° N, 43.7908° E) is is characterized by a hot, semi-arid climate and a known history of *An. stephensi* establishment, first reported in 2018 (Balkew et al., 2020). The Oromia (7.5460° N, 40.6347° E) study area comprises several districts that span an altitudinal gradient from 1500 m to 2500 m, representing a mix of agricultural, peri-urban, and highland ecosystems where the vector has been detected since 2019 (Balkew et al., 2021; Ashine et al., 2024). These sites were selected because they provide a natural experiment to evaluate the simulator's ability to capture spatial heterogeneity in vector abundance driven by temperature, urbanisation, and host availability (Degefa et al., 2025; Zhou et al., 2024).

2.3.2 Vector Biology

Anopheles stephensi is a highly adaptive urban malaria vector native to South Asia and the Middle East, now established in the Horn of Africa (Taylor et al., 2024). Its life cycle is tightly coupled to temperature, and its larvae exploit a wide range of artificial containers, especially overhead water tanks and construction pits (Thomas et al., 2016).

The temperature-dependent functions described in Section 2.2.3 were parameterised for *An. stephensi*. The coefficients (Table 2) were obtained by fitting laboratory and field data from Tuno et al. (2023), Villena et al. (2022), and Reisen and Mahmood (1980) using a Python notebook (see `species_parameters_extraction.ipynb` in the code linked to this work – Section 5). All fitted parameters are stored in the file `species_parameters.yaml`.

The host carrying capacity K uses human population density (obtained from the WorldPop raster) as a proxy for biting resources, assuming that *An. stephensi* is highly anthropophilic in urban settings (Ashine et al., 2024; Hamlet et al., 2022). Density-dependent larval mortality follows the tank capacity model of Thomas et al. (2016), where capacity is linearly scaled with water volume (maximum 500 larvae at 100 % water volume).

Table 2: Species-specific parameters for *Anopheles stephensi* used in the case study.

Parameter	Meaning	Value / coefficients	Source
Development rates			
d_E	Egg dev. (day^{-1})	$\rho = 0.0050, k = 39.208, \Delta = 2.000, \lambda = -0.855$	Reisen and Mahmood (1980)
d_L	Larva dev. (day^{-1})	$a = 2.71 \times 10^{-5}, T_0 = 5.12, T_m = 45.00, m = 1.66$	Villena et al. (2022)
d_P	Pupa dev. (day^{-1})	$\rho = 0.0051, k = 39.940, \Delta = 2.000, \lambda = -0.908$	Tuno et al. (2023)
Mortality rates [$\exp(b_1 + b_2T + b_3T^2)$]			
μ_E	Egg mort. (day^{-1})	$b_1 = 3.5729, b_2 = -0.3235, b_3 = 0.00494$	Reisen and Mahmood (1980)
μ_L	Larva mort. (day^{-1})	$b_1 = 2.0000, b_2 = -0.7395, b_3 = 0.01749$	Villena et al. (2022)
μ_P	Pupa mort. (day^{-1})	$b_1 = 5.8826, b_2 = -0.5785, b_3 = 0.00946$	Tuno et al. (2023)
μ_A	Adult mort. (day^{-1})	$b_1 = -1.4775, b_2 = -0.1377, b_3 = 0.00391$	
	Constant adult mort.	$\mu_A^{\text{const}} = 0.1198$	-
Fecundity and sex ratio			
F	Fecundity	$r_{\max} = 1.602, T_{\text{opt}} = 30.634, c = -0.00527$	Villena et al. (2022)
	Sex ratio	0.5	Assumed
Carrying capacity			
K	Host scaling factor	$1 + \ln(1 + \text{popdens})$	WorldPop Chen et al. (2020)

2.3.3 Data Sources

The simulation integrates five categories of data:

- **Static environmental rasters** – Elevation (SRTM, 30 m), building density (Global Human Settlement Layer), and human population density (WorldPop, 2020) were resampled to a common 0.001° (∼110 m) grid over the study area.
- **Climate time series** – Hourly 2 m temperature and total precipitation for 2020 were obtained from the ERA5-Land reanalysis (Copernicus Climate Change Service) and stored as NetCDF files.

- **Vector occurrence data** – Presence points for *An. stephensi* in Ethiopia were compiled from the Vector Atlas database (VectorAtlas, 2025), GBIF (Telenius, 2011), and recent entomological surveys (Balkew et al., 2020, 2021; Hawaria et al., 2023). Only records with precise coordinates and collection dates in 2018–2022 were retained.

- **Bionomic and life-history data** – Parameter values (Table 2) were extracted from peer-reviewed literature (Villena et al., 2022; Jude et al., 2023; Ghosh et al., 2025) and supplemented with expert knowledge.

All data were harmonised to the WGS-84 geographic coordinate system (EPSG:4326) and stored in a structured file hierarchy.

2.3.4 Simulation Assupmtions, Rules and Configuration

The simulation makes the following key assumptions:

1. **Temperature is the primary abiotic driver** – All development, survival, and reproduction rates respond only to ambient temperature; effects of relative humidity, rainfall, and other variables are omitted or implicitly captured through breeding site water dynamics.

2. **Host availability scales with human population density** – The effective fecundity $F \cdot K$ uses population density as a proxy for blood-meal availability, assuming that females feed preferentially on humans in urban settings (Hamlet et al., 2022).

3. **Aquatic habitats are static** – Water tank positions and their maximum water volume are fixed at initialisation; only evaporation and rainfall modify water level over time.

4. **Mosquito movement is localised** – Adults disperse via a random walk with a fixed step size (0.000 05° $\approx$ 5.5 m per tick) and a search radius of 0.005° ($\sim$550 m), reflecting limited flight range (Sayeed Quraishi et al., 1966).

5. **Resting and feeding follow simple rules** – Adults rest indoors or outdoors based on building density and temperature, and they feed when energy falls below a threshold, with success depending on local population density.

The behavioural rules (Table 3) are encoded in the YAML configuration file (`simulation.yaml`) and evaluated in priority order at each tick. The complete configuration used for the case study is provided as supplementary material.

Table 3: Behavioural rules for *An. stephensi* adults.

Condition	Action	Priority
Stage = Adult **and** energy < 0.6 **and** not resting	`feed`	8
Stage = Adult **and** energy > 0.7 **and** not gravid	`get_gravid`	7
Gravid = True **and** $T > 15\,°\mathrm{C}$	`lay_eggs`	6
$T < 10\,°\mathrm{C}$ **or** $T > 40\,°\mathrm{C}$	`die`	9
Precipitation > 0.008 m **and** build. dens. > 0.1 **and** not resting	`rest_in_building`	5
Stage = Adult **and** not resting	`move_random`	3
Otherwise	`rest`	2

The simulation was run for 180 days (six months, tick = 15 min, 17 280 ticks) with a burn-in of 5 days to allow transients to decay. Output snapshots were exported every 100 ticks and analysed using the Python notebook described in Section 2.2.6.

2.3.5 Experimental Setup

All simulations were executed on a laptop equipped with an Intel Core i5-6300U processor (2 physical cores, 4 threads, base frequency 2.40 GHz, max turbo 3.00 GHz) and 8 GB of RAM. The operating system was Linux (x86_64), and the Java Virtual Machine (JVM) was configured with a maximum heap size of 1950 MB (approximately 80 % of physical memory). The simulation engine used a thread pool with three worker threads (one less than the total number of logical threads) to avoid oversubscription. The spatial registry implemented a lazy grid with a cell size of 0.001° ($\approx$110 m). Each simulation ran for 180 simulated days (17 280 ticks of 15 min). Output snapshots were exported every 100 ticks. The analysis notebook (Python 3.10) was executed on the same hardware.

3 Results

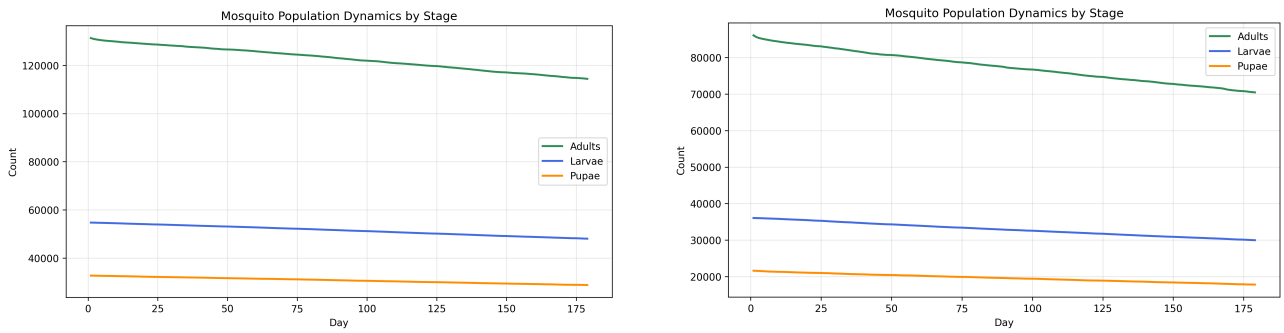
The simulation was executed for a duration of six months (180 days, comprising 17 280 ticks of 15 min each) over the contrasting study areas of Somali and Oromia in Ethiopia. The model utilised the temperature-dependent lifecycle parameters specifically fitted for *Anopheles stephensi*. Output snapshots were processed to evaluate population stability, environmental response, and spatial suitability.

3.1 Population Dynamics and Lifecycle Validation

In the Oromia region, the mosquito population reached a near-steady state after approximately 10 days of simulation, with adult abundance stabilising at approximately 85 000 individuals, while larval and pupal counts fluctuated around 36 000 and 21 600, respectively (Figure 2). Analysis of the six-month time series revealed that the daily adult count maintained a weak positive correlation with temperature (Pearson's $r = 0.119$, $p = 0.12$) and a stronger positive correlation with precipitation ($r = 0.618$, $p = 1.56 \times 10^{-19}$).

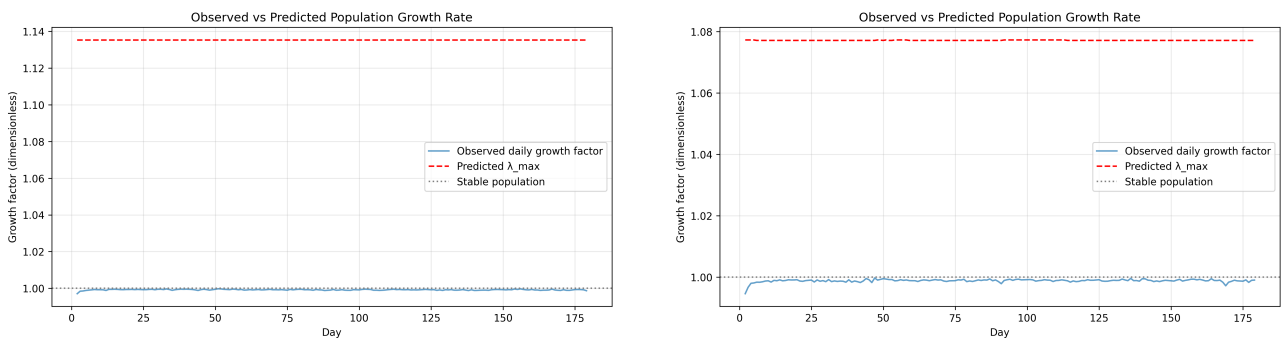
In the Somali region, the adult count started at approximately 131 000 and showed a slow but consistent increase, with larvae and pupae averaging 54 700 and 32 700, respectively. Temperature remained nearly constant (26.3(1) °C) throughout the simulation, leading to an undefined correlation coefficient. Precipitation exhibited a moderate negative correlation with adult abundance ($r = -0.477$, $p = 3.8 \times 10^{-11}$). These results indicate that under the simulated conditions, intrinsic density-dependent regulation and the availability of stable aquatic habitats dominated over short-term climatic fluctuations in both regions.

Theoretical stage durations were calculated based on the mean observed temperatures: at 21.5 °C (Oromia) the larval and pupal stages lasted 15.7 d and 4.8 d, respectively; at 26.3 °C (Somali) the corresponding durations were 11.4 d and 4.3 d. The dominant eigenvalue ($\lambda_{\max}$) of the Metzler projection matrix, averaged over the full simulation period, was 1.077 for Oromia and 1.135 for Somali. These values indicate a slowly growing population approaching equilibrium, consistent with the established status of *An. stephensi* in urban Ethiopia (Balkew et al., 2021). Figure 3 illustrates the observed daily adult growth factor alongside the predicted $\lambda_{\max}$.



(a) Somali (b) Oromia

Figure 2: Daily counts of adult, larval, and pupal *An. stephensi* over the six-month simulation. Population levels stabilised after an initial burn-in period of about 10 days.

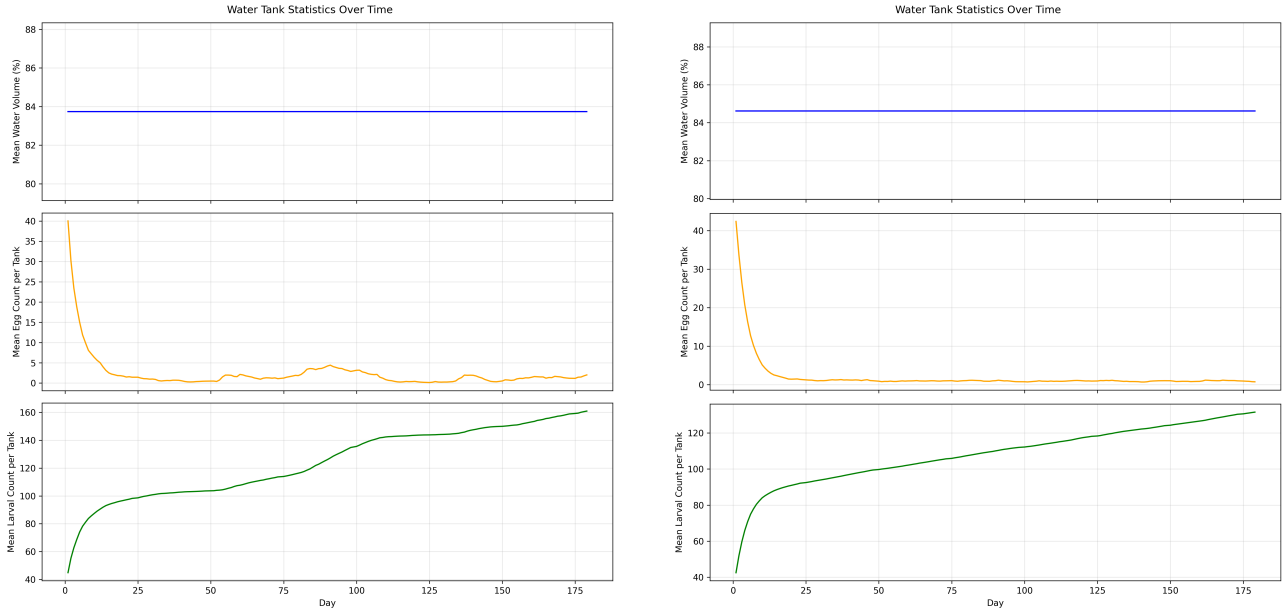


(a) Somali (b) Oromia

Figure 3: Observed daily adult growth factor versus predicted $\lambda_{\max}$ (red dashed line). The average predicted growth factor closely matches the observed stability in both regions.

3.2 Water Tank Statistics

The Oromia simulation included 474 water tanks, while the Somali simulation included 42 tanks. In both regions, mean water volume remained highly stable (84.6(1) % for Oromia, 83.7 % for Somali), as evaporation losses were balanced by periodic precipitation events (Figure 4). The reciprocal trend between egg and larval counts per tank confirmed active hatching and stage transition within these permanent aquatic habitats.

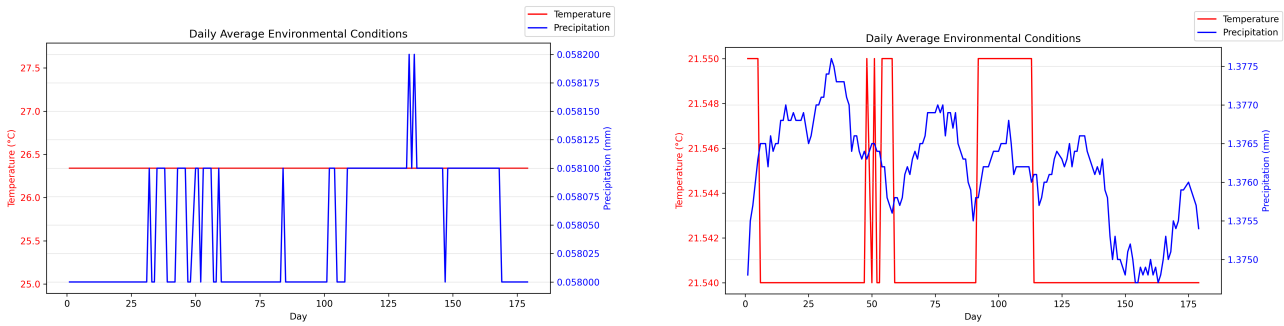


(a) Somali (b) Oromia

Figure 4: Water tank statistics: (top) mean water volume; (middle) mean eggs per tank; (bottom) mean larvae per tank.

3.3 Environmental Correlation

Daily average temperatures ranged between 20.7 °C to 23.8 °C in Oromia and remained almost constant at 26.34 °C in Somali. Precipitation was low and relatively constant in Somali ($\sim 0.058\,\mathrm{mm\,d^{-1}}$), while Oromia experienced higher and more variable precipitation (mean $1.35\,\mathrm{mm\,d^{-1}}$; Figure 5). The low variability in these drivers, especially in Somali, explains the dominance of endogenous population processes over short-term climatic forcing in this specific case study.



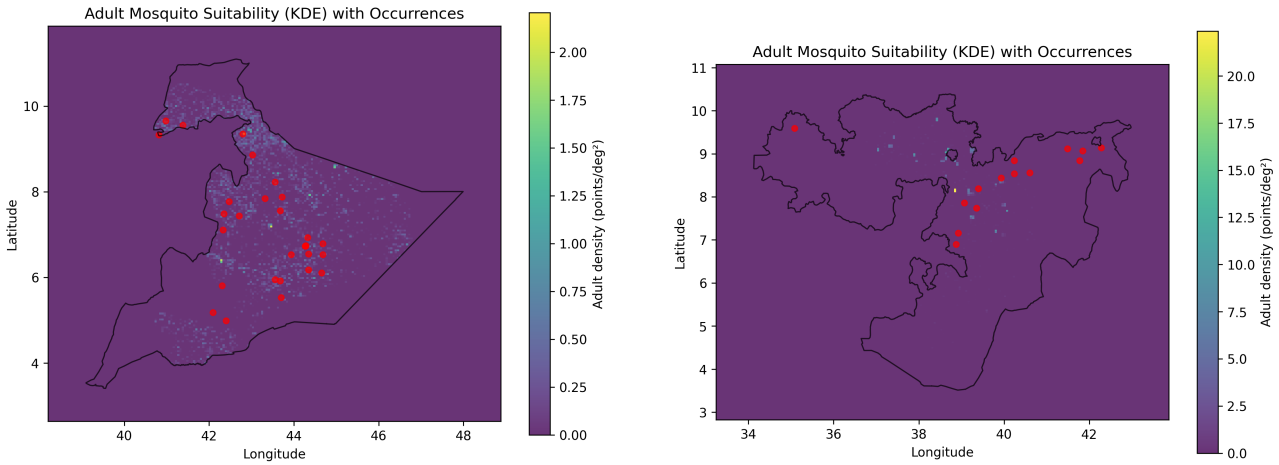
(a) Somali (b) Oromia

Figure 5: Daily average environmental conditions (temperature in red, precipitation in blue) throughout the simulation.

3.4 Spatial Suitability and Validation

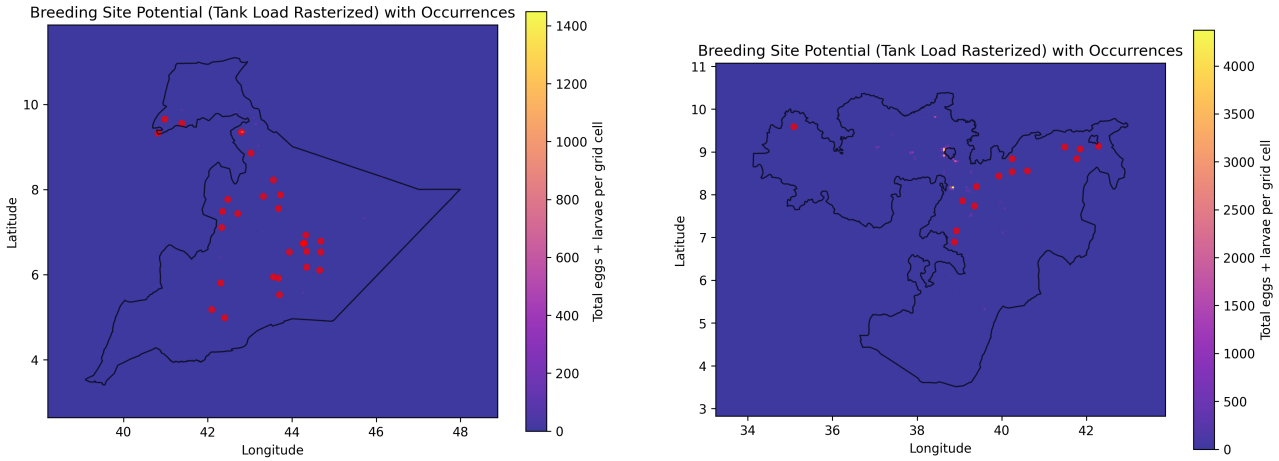
Spatial suitability maps were generated using Kernel Density Estimation (KDE) with a bandwidth of 0.002° ($\sim 220\,\mathrm{m}$). As shown in Figure 6, adult mosquito density was highest in urban cores and areas with significant building density. Validation against independent occurrence points yielded a continuous Boyce index of 0.504

for Oromia and 0.782 for Somali. The proportion of occurrence points falling within the top 20% of the adult suitability raster was 35.7 % for Oromia and 36.7 % for Somali, confirming the model's moderate to good predictive power for invasive vector establishment. The final breeding site potential, derived from the distribution of eggs and larvae in water tanks, is presented in Figure 7.



(a) Somali (b) Oromia

Figure 6: Adult mosquito suitability maps (KDE) overlaid with occurrence records (red dots). High density correlates with urban infrastructure.



(a) Somali (b) Oromia

Figure 7: Breeding site potential rasterized from final egg and larval counts in water tanks, overlaid with occurrence records (red dots).

3.5 Performance Metrics

Table 4 summarises the computational and agent statistics for both study sites. Both simulations completed the full 180-day period (17 280 ticks) without errors. The Oromia run had a slightly lower total runtime (2253 s vs. 2439 s) and a slightly higher effective frame rate (8.2 FPS) despite having a larger spatial grid (60.9 million cells). Memory consumption was significantly higher for the Somali simulation (1305 MB vs. 650 MB) due to the larger number of agents (191 092 vs. 118 614).

Both runs stayed well below the available memory limit, confirming that a standard laptop can execute the model for region-scale simulations. The average tick time remained below 125 ms, enabling the simulation of six months of vector population dynamics in less than one hour of wall-clock time on consumer hardware.

Table 4: Performance metrics and agent counts for the two simulation runs.

Metric	Somali	Oromia
Total runtime (s)	2439.1	2252.9
Average tick time (ms)	124.5	122.6
Effective FPS	8.0	8.2
Peak memory usage (MB)	1305	650
Memory utilisation (% of max)	69.5	33.3
Total agents (end)	191092	118614
Mosquitoes	191050	118140
Water tanks	42	474
Rule evaluations	6716	6603
Actions executed	18512932	19650028
Spatial registry — inserts	224614	194738
Spatial registry — updates	151	138
Max agents per spatial cell	513	319

4 Discussion

The simulation results demonstrate that the agent-based model successfully reproduces population dynamics for *Anopheles stephensi* across two distinct Ethiopian ecological settings. In the Oromia *woredas*, the population maintained a near-equilibrium state with an average predicted $\lambda_{\max}$ of 1.077, whereas in Somali, the model predicted a more robust growth potential ($\lambda_{\max} = 1.135$). These findings align with field reports that identify Somali as a primary hotspot for the species' establishment in urban Ethiopia (Balkew et al., 2021). The theoretical stage durations – 15.7 d for larvae and 4.8 d for pupae in Oromia (21.5 °C) and 11.4 d and 4.3 d, respectively, in the warmer Somali region (26.3 °C) – fall within the ranges reported in laboratory studies (Jude et al., 2023), supporting the validity of our temperature-dependent parameterisation.

A key finding is the site-specific response to environmental drivers. In Oromia, adult abundance showed a weak positive correlation with temperature ($r = 0.119$, $p = 0.12$) and a stronger positive correlation with precipitation ($r = 0.618$, $p = 1.6 \times 10^{-19}$), suggesting that precipitation may enhance larval habitat availability and adult survivorship. In Somali, where temperature remained almost constant, a moderate negative correlation was observed between adult counts and precipitation ($r = -0.477$, $p = 3.8 \times 10^{-11}$), potentially reflecting a "wash-out" effect or a shift in habitat availability during high-rainfall events, a phenomenon documented for urban container-breeding mosquitoes elsewhere (Thomas et al., 2016). These contrasting responses underscore the importance of local hydro-climatic contexts when modelling *An. stephensi* stability.

The spatial suitability maps generated via KDE correctly highlighted urban cores and high-building-density areas as the most suitable habitats. Validation against independent occurrence points yielded a Continuous Boyce Index (CBI) of 0.504 for Oromia and 0.782 for Somali, indicating moderate to high predictive performance for both regions. The proportion of occurrence points falling within the top 20 % of predicted suitability areas was 35.7 % for Oromia and 36.7 % for Somali, confirming that the model effectively captures high-risk zones across different spatial settings.

The performance analysis (Table 4) demonstrates that the simulation framework is computationally efficient and can run on modest hardware. Both six-month simulations completed in less than 2450 s (approximately 40 min) on a standard laptop, with average tick times below 125 ms. Peak memory usage remained 1317 MB for the larger Somali simulation (191k agents) and 650 MB for the Oromia simulation (119k agents), well under the 2 GB heap limit. Notably, the Oromia simulation handled a very sparse spatial grid of 60.9 million cells (due to the larger study area) with only 474 water tanks, while the Somali simulation used a grid of 68.5 million cells with 42 tanks. The lazy spatial registry kept performance acceptable despite the vast number of empty cells, with rule evaluations and actions executed remaining within practical limits. These characteristics are critical for deployment in sub-Saharan African public health institutions, where access to high-performance computing clusters is often limited.

This opens the door for routine use by national malaria control programs and regional health observatories, enabling them to run scenario analyses, evaluate intervention strategies, and generate risk maps using only standard office computers. Future work will focus on reducing the memory footprint further and providing user-friendly interfaces to lower the technical barrier for public health officers in endemic areas.

Several limitations should be acknowledged. The current movement rules utilise a random walk with a fixed step size, which may not fully capture complex host-seeking behaviours toward specific resource gradients. Furthermore, the model assumes human population density is the sole proxy for blood-meal availability. Given that *An. stephensi* exhibits plastic feeding behaviour, including zoophily, a multi-host formulation might improve

fecundity estimates in peri-urban areas. Despite these limitations, we believe the framework provides a valuable tool for designing targeted vector control interventions. The YAML-driven parameterisation allows for rapid adaptation to other regions or species. Future research should extend these simulations to multi-year runs to evaluate the impact of long-term climate variability and the effectiveness of larval source management, such as the systematic covering of water tanks, on population suppression.

5 Conclusion

This study presented an agent–based modelling framework for simulating the population dynamics of mosquitoes in low–resource settings, with a case study on *Anopheles stephensi* in two Ethiopian regions (Somali and Oromia). The model successfully integrated temperature–dependent life–history traits with spatially explicit environmental data and a stage–structured Metzler matrix. The framework was applied to Somali and Oromia using six–month simulations driven by ERA5–Land data.

The results demonstrated that the simulated populations reached near–equilibrium or slowly growing states consistent with field observations of established vectors. In Oromia, the average predicted $\lambda_{\max}$ was 1.077, and the adult population stabilised at approximately 85 000 individuals. In Somali, a higher growth potential ($\lambda_{\max} = 1.135$) was observed, with adult numbers starting at 131 000 and continuing a slow increase. The model identified significant regional differences in climatic sensitivities: Oromia showed positive correlations with both temperature and precipitation, while Somali (with nearly constant temperature) exhibited a moderate negative correlation with precipitation. Validation against independent occurrence data yielded Continuous Boyce Indices of 0.504 for Oromia and 0.782 for Somali, with 35.7 % and 36.7 % of occurrence points falling within the top 20 % of suitability, respectively. These results confirm the model's ability to identify urban hotspots even in data–poor environments.

Computational performance was adequate for routine use: both six–month simulations completed in less than 2450 s on standard laptop hardware, with average tick times below 125 ms and peak memory usage under 1.4 GB. While limitations regarding adult dispersal and host–seeking behaviour remain, the framework provides a generic, reusable, and easily parameterisable tool for public health decision–making. By linking life–cycle biology with spatial infrastructure data, the model could support evidence–based planning of larval source management and urban malaria control programmes in recently invaded territories.

Data availability

The code of the simulator and the data used for the case study are all available in a github directory: https://github.com/Vansnoden/monadvsim.

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